

# The Accuracy of Supervised Machine Learning Algorithms in Predicting Cardiovascular Disease

Jatin Gupta  
CodeHand.org  
England, United Kingdom  
jatin@ieee.org

**Abstract**—In recent years, the application of machine learning models is becoming more prevalent in identifying, predicting, and diagnosing cardiovascular disease (CVD). CVD is known to be the top cause of death globally and about 17.9 million lives are lost each year due to CVD, according to the World Health Organisation. Studying the accuracy of machine learning algorithms is significant in identifying a predictive model which is most efficient in diagnosing cardiovascular diseases in patients, on the basis of their health history. In this research, the CVD dataset (accessible from Kaggle), which has 70,000 records of different patients' case histories and 11 attributes for each patient, has been used. With the aim of identifying the most efficient model, different supervised machine learning algorithms are built and compared, implementing the decision tree classifier, the random forest classifier, K-nearest neighbors, and support vector machine classifier using Python 3.7. The results showed that the support vector machine predictive model gives a better result than the other models, in terms of the accuracy, Area Under the Curve (AUC) score and F1 score. The contribution of this research will be the application of a predictive model through which CVD can be identified, diagnosed, and treated at early stages.

**Keywords**—Supervised machine learning algorithms, cardiovascular disease, support vector machine classifier, K-nearest neighbor classifier

## I. INTRODUCTION

Machine learning is a sub-field of artificial intelligence (AI). It involves the creation of algorithms which can learn automatically from data available, with minimal human involvement [1]. There are different kinds of machine learning algorithms which are capable of processing huge amounts of data, transforming this data into useful knowledge, which in turn can be used to make intelligent decisions [2]. Since the early 2000s, interest in machine learning has been growing and machine learning algorithms are being applied in healthcare, such as in cancer prediction, early diagnosis of different diseases, and cardiovascular diseases (CVD) risk prediction [3].

The term 'cardiovascular disease' covers a range of diseases which affect the heart or the blood vessels, such as coronary heart disease [4]. CVD are the top causes of death globally. An estimated 17.9 million lives are lost each year (31% of all deaths worldwide) due to CVD [5]. Thus, it is important to study heart related diseases and understand how the application of machine learning can transform available data into useful knowledge that can help in making significant assessments on the health of patients.

The use and implementation of machine learning algorithms is becoming increasingly prevalent in the diagnosis of CVD, due to the availability of huge amounts of health-related data. Machine learning algorithms are applied in healthcare systems in order to assist with the accurate detection and diagnosis of CVD in patients. This would

allow for more specialized and early treatment of those who are predicted to suffer with CVD [6].

In this research, supervised machine learning algorithms are implemented to identify and build an optimum predictive model on CVD. The aim of this research is to explore and compare supervised machine learning algorithms (decision tree classifier, random forest classifier, K-nearest neighbors classifier, and support vector machine classifier) to find out which predictive model is the most efficient in predicting CVD in patients, on the basis of their healthcare history.

## II. LITERATURE REVIEW

Cardiovascular disease (CVD) has been listed as one of the prominent causes of death in all countries. It is known that about half of the deaths in certain developed countries (such as the United States) are due to heart diseases [7]. According to WHO statistics, it is being estimated that in India, 24% deaths are because of non-communicable diseases caused by heart ailments, and about one-third of all global deaths are caused due to heart disorders [5]. Thus, CVD is known to be one of the largest diseases in terms of impacting people and in causing complications which can prove to be fatal [8].

According to researchers [9], the number of fatalities caused by CVD can be reduced if physicians are able to identify and diagnose CVD in patients in the early stages of the disease. If diagnosed early then it can be treated early, avoiding complications which may appear in the advanced stages of CVD. Thus, many researchers and practitioners [10-11] are experimenting and implementing machine learning techniques, such as neural networks, support vector machines, decision trees, naïve bayes, random forest, logical regression, AutoPrognosis, and gradient boosting, to anticipate cardiovascular diseases.

In paper [12], researchers mentioned that identifying people who are at risk of CVD is significant in preventing cardiology related health problems. However, the authors mentioned that risk prediction models recommended by clinical guidelines may not show good performances as they are based upon a limited number of predictors. Thus, they experimented whether the use of machine learning models would improve the performance of risk predictions as machine learning techniques are able of learning from the complex interactions of risk variables. The data for this study consisted of 423604 people without CVD in UK Biobank and developed a machine learning based risk predicting model from 473 variables available [12]. Their machine learning based model was derived utilising AutoPrognosis. This is an algorithmic tool that selects and tunes feature processing, classification, and calibration algorithms. The results of the model derived were then compared with the Framingham score (common CVD risk factors), Cox proportional hazards (based on known risk factors such as age, gender, systolic blood pressure etc.) and a Cox PH

model on all of the 473 variables. The assessment of the predictive performances was carried out using area under the receiver operating characteristic curve (AUC-ROC). The results highlighted that the author's AutoPrognosis model had an improved risk prediction as compared to other models (Framingham score, Cox PH model with usual risk factors and Cox PH model with UK Biobank variables). The study concluded that the AutoPrognosis model improved the accuracy of CVD risk prediction population of UK Biobank [12].

Studies on the application of machine learning techniques in healthcare have been conducted in various parts of the world. Some researchers believe that it is important to experiment with machine learning techniques with CVD datasets available in different countries. For example, researchers [13], explored the Korean National Health Insurance Service-National Health Sample Cohort (KNHSC) data to predict heart disease by studying the features of machine learning and the datasets. In this study, the accuracy of different machine learning methods was assessed to predict the two year and ten year risk of heart diseases like heart failure, coronary artery disease, and strokes. In this study, machine learning based prediction models were developed through deep neural networks, random forests, and logistic regression, and then were validated using the metrics F1 score, AUC-ROC, and precision-recall curves. The results of the study showed that machine learning models are very useful in predicting ten year CVD risks.

In research [14], it is stated that in various research projects, machine learning algorithms are used to predict CVD, but these methods have limitations due to the ambiguity of the predictable model structure, and inadequate data.

This section on literature review highlights the importance of utilising machine learning techniques through using datasets that are available in different countries. The literature review also confirms that there are limited studies on the implementation of supervised machine learning algorithms using CVD dataset. Thus, in this research, supervised machine learning algorithms are implemented and compared to identify a predictive model which is the most efficient in identifying CVD.

### III. METHODOLOGY

#### A. Data and its Attributes

In this section, the data of the cardiovascular disease dataset is explored to gain an understanding about the variables which impact the target feature (whether the patient had any CVD).

In this research, the CVD dataset is obtained from Kaggle (sourced from [15]) which has 70,000 records of different patients' case histories. In this dataset, age, weight, height, gender, blood pressure, cholesterol, glucose, smoking, alcohol and physical activity are the attributes.

The target feature is cardiovascular disease. This has variable name 'cardio' and data type binary, informing if a person has cardiovascular disease or not. The target variable "cardio" is 1 when a patient has cardiovascular disease, and if the patient does not have cardiovascular disease, it is 0.

Table 1 presents a description of the data. In the CVD dataset used in this research, three kinds of input features are recorded. Objective data provides factual information, examination data is recorded based upon the results of a medical examination, and subjective information is taken from the patients.

Table 1. Data description

| Features                 | Objective | Subjective | Examination | Variable    | Value Type                                      |
|--------------------------|-----------|------------|-------------|-------------|-------------------------------------------------|
| Age                      | ✓         |            |             | Age         | int (days)                                      |
| Height                   | ✓         |            |             | Height      | int (cm)                                        |
| Weight                   | ✓         |            |             | Weight      | float (kg)                                      |
| Gender                   | ✓         |            |             | Gender      | categorical                                     |
| Systolic blood pressure  |           |            | ✓           | ap_hi       | int                                             |
| Diastolic blood pressure |           |            | ✓           | ap_lo       | int                                             |
| Cholesterol              |           |            | ✓           | cholesterol | 1-normal, 2- above normal, 3- well above normal |
| Glucose                  |           |            | ✓           | gluc        | 1-normal, 2- above normal, 3- well above normal |
| Smoking                  |           | ✓          |             | smoke       | binary                                          |
| Alcohol intake           |           | ✓          |             | alco        | binary                                          |
| Physical activity        |           | ✓          |             | active      | binary                                          |

#### B. Data Processing

During data pre-processing, it was found that the CVD dataset has 1451 outliers. This can create a misleading statistic during data processing, hence should be removed. After removing the outliers, 68549 data records were considered. The models were trained on 47984 data records and tested on 20565 records, in a 70:30 ratio. The predictive model was built in the Python 3.7 programming language.

The datasets are represented as below:

Complete dataset  $D = \{D1, D2, D3, D4 \dots D70000\}$

Training dataset as  $\text{Train} = \{D1, D2, D3, D4 \dots D47984\}$

Test data as  $\text{Test} = \{D47985, D47986 \dots D70000\}$

In summary, there are two stages of applying the machine learning techniques. At first, part of the available data is used to train the model, the training set. Then, using the trained model, the remainder of the data is used to test the model, the testing set. This helps to evaluate the accuracy of the model [2].

#### C. Supervised Machine Learning Algorithms

A supervised machine learning algorithm understands the function which transforms an input to an output. It learns this function from analysing the training data, and can then produce outputs for unseen inputs.

#### a) Decision Tree Classifier:

The Decision Tree Classifier (DTC) is a supervised machine learning model. It can be compared to or thought of as a flowchart, in that each node (representing one attribute) leads into two or more sub-nodes. Similar to how flowcharts assist in decision making processes through following simple statements, a DTC predicts the value of a variable through the decision rules it has learnt or inferred from the training data [16].

A DTC mainly consists of nodes (which test the value of a particular attribute), edges (connecting a node to another node), and leaf nodes (which predict the final value) [16].

The DTC algorithm runs as follows [17]:

Firstly, locate the tree root and divide the data based on the feature which results in the largest reduction in uncertainty of the final result. Then, this procedure is repeated until the samples at each leaf node are of the same class and a limit must be placed on the depth of the tree to prevent overfitting of the data.

#### b) Random Forest Classifier:

The Random Forest Classifier (RFC) is an ensemble and supervised machine learning model used for classification and regression. RFCs use multiple Decision Trees and combine these in order to receive a higher accuracy prediction.

For example, five Decision Trees, which output a 0 or a 1 based upon some data given, can be part of a Random Forest Classifier. The testing data is run on the Random Forest Classifier, and each of the five Decision Trees will give us a 0 or 1 as output. The overall prediction of the model is the value which the majority of the Decision Trees chose. So, if four decision trees outputted 1, and one decision trees outputted 0, the Random Forest Classifier would output a 1 [18].

#### c) K-Nearest Neighbors Classifier:

According to author [19], the K-Nearest-Neighbors (KNN) Classifier is a supervised machine learning model which can be used for both classification and regression. In KNNs, the output depends on whether classification or regression is being used. K is simply any integer number.

If classification is being used, then the output of a particular piece of data will be the most common output of the k nearest neighbors, which is the mode of the outputs. In regression, the output value will be the mean of these k nearest neighbors [20].

As the value of k increases from 1, the predictions begin to be more accurate. However, this trend may not continue and there is usually an optimum value of k in the KNN algorithm.

The KNN algorithm is advantageous in that a model does not need to be built and that the algorithm has many applications. However, the speed of the algorithm decreases as k increases.

The KNN algorithm can be visualised using the figure 1.

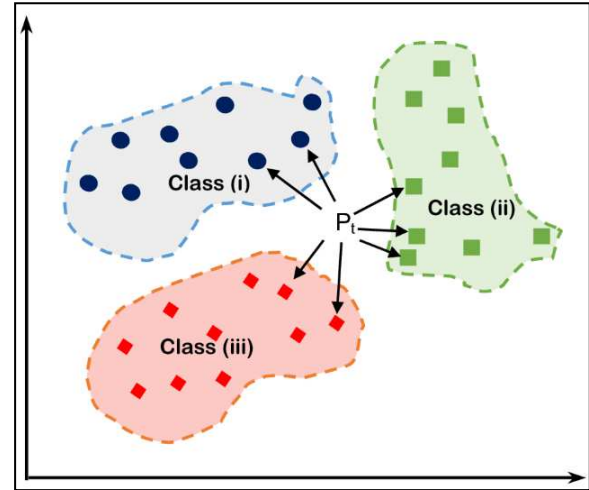


Fig. 1. KNN algorithm [20]

The KNN algorithm has been trained on data, and similar to this project, using regression. To predict the output value of new data, the mean of the output values of the data from the k nearest points to the new data point is taken. For example, if  $k = 5$ , the output values of the five closest points to the new data point will be summed and divided by 5, and this value given as an output value to the new point added to it.

#### d) Support Vector Machine:

The Support Vector Machine (SVM) is a supervised machine learning model which can be used for both classification and regression. The SVM algorithm (figure 2) determines a hyperplane (boundary) in N-dimensions that can classify the data points. Here,  $N$  = number of attributes of data. If the data has two attributes, then the hyperplane is simply a line separating the data points. If the data has three attributes, then the hyperplane is a plane separating the data points. Therefore, data points which lie on different sides of the hyperplane would correspond to different classes [21].

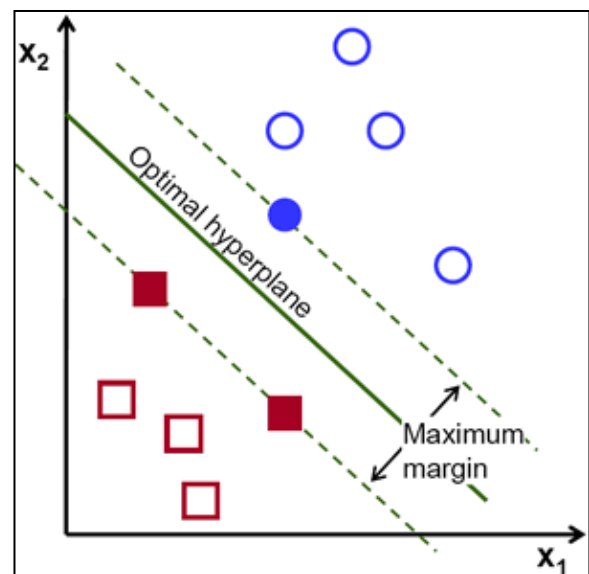


Fig. 2. Possible hyperplanes [22]

When separating data points, it is possible that there are several options for a hyperplane which can be chosen. However, with SVM, it must be ensured that the plane found has the maximum distance between points of the classes.

#### IV. RESULTS AND DISCUSSION

In this project, the decision tree classifier, random forest classifier, K-nearest neighbors classifier, and support vector machine classifier were implemented. The results for these models are presented in this section.

In figures 3, 4 and 5, the y-axis label represents a particular attribute, or feature, and the x-axis label has a 0 and a 1. The 0 represents cases in patients where there was no CVD, and the 1 represents cases in patients where there was CVD. These figures can be used to develop an understanding of the attributes and factors that are associated with CVD in patients.

If the diagram shows similar patterns for a particular attribute between the patients who had CVD and the patient who did not have CVD, it can be inferred that the particular attribute does not affect the patient's chance of developing CVD.

The figures were generated in addition to the machine learning models which were built in Python 3.

##### A. Age and Gender Variables

Figure 3 details the age and gender specification, with respect to whether CVD occurred or did not occur in patients from the dataset.

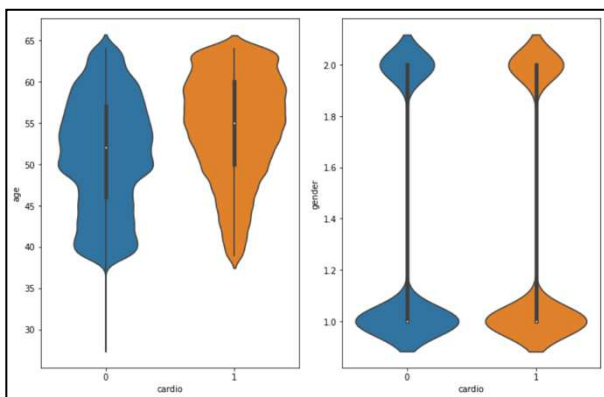


Fig. 3. Age and gender

In the above violin plot, by comparing the probability density at different points, it can be seen that those with CVD tend to be slightly older than those with no CVD, in addition to those with CVD having a higher mean age. However, gender does not affect CVD in patients.

##### B. Weight and Height Variables

The violin plot (in figure 4) indicates that the distribution of heights for patients with CVD is similar to the distribution of heights for patients with no CVD. However, the distribution of weights for patients with CVD is slightly higher than the distribution of weights for patients with no CVD.

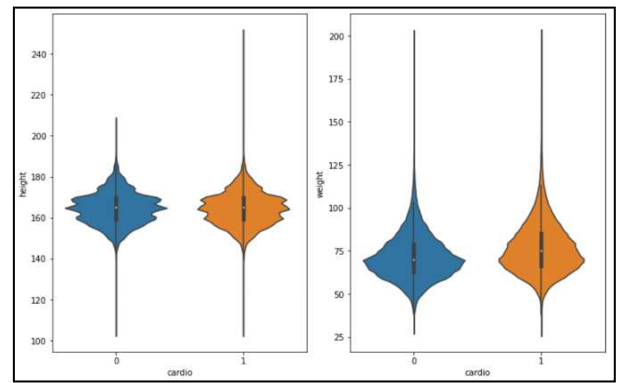


Fig. 4. Statistics of weight and height variable

The left of the figure shows how the height attribute varies with CVD in patients. It can be deduced that the height attribute is associated with CVD in patients to a low extent.

However, in the right of the figure, it can be seen that the weight of the patient is a significant attribute in determining whether a patient has CVD. This represents that as the weight of a patient increases, the chances of them developing CVD is likely to increase.

##### C. Systolic Blood Pressure and Diastolic Blood Pressure

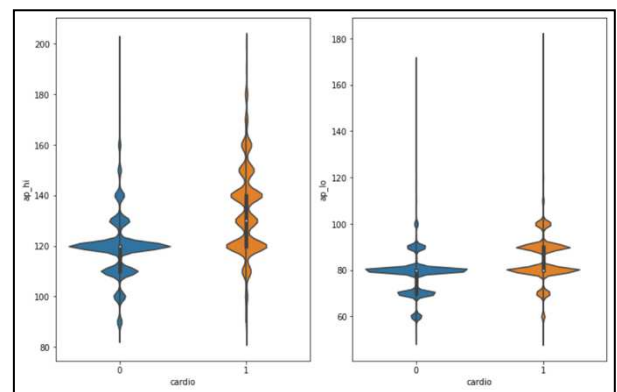


Fig. 5. Systolic and diastolic blood pressure

Figure 5 details two diagrams. On the left is ap\_hi (systolic blood pressure of the patient), and on the right is ap\_lo (diastolic blood pressure of the patient).

In this violin plot, it can be seen that those with CVD have a higher mean ap\_hi than those with no CVD. Whilst the mean of the ap\_lo in patients with and without CVD is similar, a greater number of patients with CVD had a higher ap\_lo than those with no CVD.

### D. Relationship Between Variables

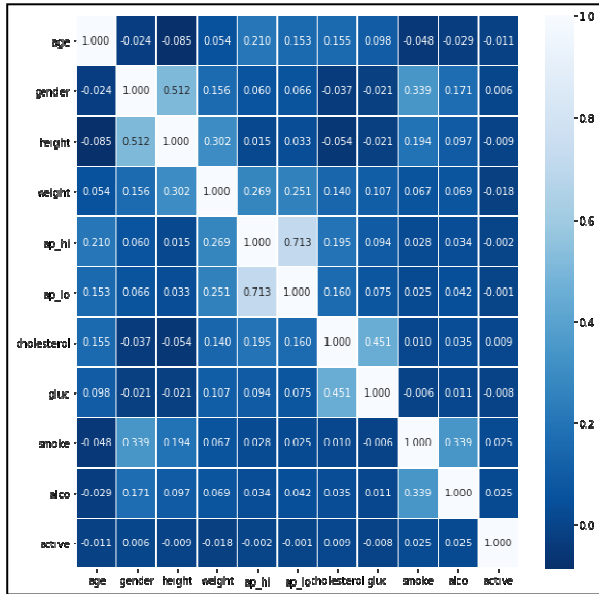


Fig. 6. Correlation matrix between data attributes

The above heat map shows the correlation matrix and represents the relationship between attributes in the dataset. The associations of weight, smoking, cholesterol, glucose, and systolic blood pressure show a strong association and have a significant impact on CVD in patients.

### E. Accuracy of Supervised Machine Learning Algorithms

The accuracy of the predictive model shows that the decision tree classifier accuracy is 63%, the random forest classifier accuracy is 70%, the KNN classifier accuracy is 72%, and the SVM classifier accuracy is 73%.

Table 2. Accuracy score of models

| S. No. | Supervised Machine Learning Model Classifier | Accuracy | AUC score |
|--------|----------------------------------------------|----------|-----------|
| 1      | Support Vector Machine Classifier            | 0.732215 | 0.79      |
| 2      | Decision Tree Classifier                     | 0.628933 | 0.63      |
| 3      | K-Nearest Neighbor Classifier                | 0.723705 | 0.79      |
| 4      | Random Forest Classifier                     | 0.703428 | 0.76      |

Figure 7 shows the ROC curve of the different predictive models built in this research.

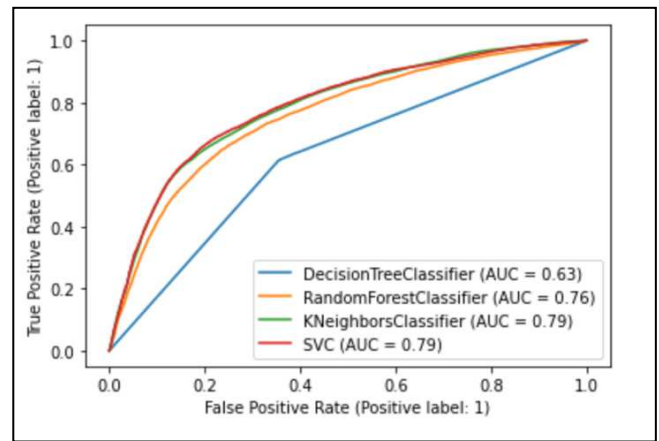


Fig. 7. ROC of different predictive model accuracy

From table 2 and figure 7, it can be seen that the SVM and KNN classifiers are performing better than other models.

For the two models with the highest accuracy and AUC score, further scores were calculated for them (table 3). This research verified that SVM gives a better result than other models, in terms of Accuracy score, AUC score and F1 score.

Table 3. Comparison of scores between support vector machine and K-nearest neighbor

| S. No. | Supervised Machine Learning Model Classifier | Precision | Recall score | F1 score |
|--------|----------------------------------------------|-----------|--------------|----------|
| 1      | Support Vector Machine Classifier            | 0.79178   | 0.71049      | 0.74894  |
| 2      | K-Nearest Neighbor Classifier                | 0.76248   | 0.71082      | 0.73574  |

### V. CONCLUSION

The decision tree classifier, the random forest classifier, K-nearest neighbors, and support vector machine classifiers were built and implemented, in order to find out which predictive model is the most efficient in predicting cardiovascular disease in patients on the basis of their health history. The testing of the data shows that the decision tree classifier has an accuracy of 63%, the random forest classifier has an accuracy of 70%, the K-nearest neighbors classifier has an accuracy of 72%, and the support vector machine classifier has an accuracy of 73%. This shows that the support vector machine classifier and K-nearest neighbors classifiers performed better than the other models. It can be verified that the support vector machine predictive model gives a better result than other models, in terms of accuracy score, AUC score, and F1 score.

This project carried out an extensive literature review, providing a strong background for this research study, in order to formulate the research aim. The proposed predictive support vector machine model is the most accurate, due to its AUC score and F1 score. Therefore, the use of the support vector machine classifier built and implemented in this research to predict CVD in patients is recommended.

## REFERENCES

- [1] S. E. Awan, M. Bennamoun, F. Sohel, F. M. Sanfilippo, and G. Dwivedi (2019). Machine learning-based prediction of heart failure readmission or death: implications of choosing the right model and the right metrics. *ESC Heart Failure*, 6(2), pp. 428–435.
- [2] A. C. Dimopoulos, M. Nikolaidou, and F. F. Caballero (2018). Machine learning methodologies versus cardiovascular risk scores, in predicting disease risk. *BMC Medical Research Methodology* 18, p. 179.
- [3] W. J. Yanwei, Y. Gao, and Z. Zhao (2007). Combination data mining methods with new medical data to predicting outcome of coronary heart disease. *International conference on Convergence Information Technology*. pp. 868–72.
- [4] D. Chicco and G. Jurman (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making* 20, Article 16.
- [5] World Health Organization (2021). Available at [https://www.who.int/health-topics/cardiovascular-diseases/#tab=tab\\_1](https://www.who.int/health-topics/cardiovascular-diseases/#tab=tab_1). Accessed on 15<sup>th</sup> April 2021.
- [6] A. K. Gárate-Escamila, A. Hajjam El Hassani and E. Andrès (2020). Classification models for heart disease prediction using feature selection and PCA, *Informatics in Medicine Unlocked* 19.
- [7] K. Mc Namara, H. Alzubaidi and J. K. Jackson (2019). Cardiovascular disease as a leading cause of death: how are pharmacists getting involved? *Integrated Pharmacy Research and Practice*, 8, pp. 1–11. doi: 10.2147/IPRP.S133088. PMID: 30788283; PMCID: PMC6366352.
- [8] S. S. Virani, A. Alonso, E. J. Benjamin, C. W. Callaway (2020). Heart disease and stroke statistics—2020 update: A report, *Circulation* 141 (9), p. 139.
- [9] K. G. Dinesh, K. D. Arumugaraj, Santhosh and Mareeswari (2018). Prediction of cardiovascular disease Using Machine Learning Algorithms, *International Conference on Current Trends towards Converging Technologies (ICCTCT)*, pp. 1-7.
- [10] K. Vanisree and J. Singaraju (2011). Decision support system for congenital heart disease diagnosis based on signs and symptoms using neural networks. *International Journal of Computer Applications* 19(6). 10.5120/2368-3115.
- [11] S. Ghumbre and A. Ghatol (2012). Heart Disease Diagnosis Using Machine Learning Algorithm. 10.1007/978-3-642-27443-5\_25.
- [12] A. M. Alaa, T. Bolton, Di Angelantonio, J. H. F. Rudd and M. Schaar (2019). Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK Biobank participants. *PLoS One*.14(5).
- [13] G. Joo, Y. Song, and J. Park (2020). Clinical Implication of machine learning in predicting the occurrence of cardiovascular disease using big data (Nationwide Cohort Data in Korea)," in *IEEE Access*, 8, pp. 157643-157653.
- [14] M. Singh, M. M. Levi, P. Joanis, and V. K. Mago (2016). Creating a predictive model for heart disease using Structural Equation Model & Fuzzy Cognitive Map. *IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*.
- [15] Kaggle.com (2021). Available at <https://www.kaggle.com/sulianova/cardiovascular-disease-dataset>. Accessed on 1<sup>st</sup> May 2021.
- [16] S. S. Shai and B. D. Shai (2014). 18. Decision Trees. *Understanding machine learning*. Cambridge University Press.
- [17] A. Chakure (2019). An introduction to Decision Tree Classifier, Available at <https://medium.com/swlh/decision-tree-classification-de64fc4d5aac>. Accessed on 10<sup>th</sup> May 2021.
- [18] N. Donges (2021). A complete guide to the random forest algorithm. Available at <https://builtin.com/data-science/random-forest-algorithm>. Accessed on 20<sup>th</sup> May 2021.
- [19] P. Cunningham and S. J. Delany (2020). k-Nearest neighbor classifiers: 2nd Edition. Available at <https://arxiv.org/abs/2004.04523>. Accessed on 15<sup>th</sup> May 2021.
- [20] D. M. Atallah, M. Badawy, and A. El-Sayed (2019). Intelligent feature selection with modified K-nearest neighbor for kidney transplantation prediction. *SN Applied Science* 1, 1297.
- [21] D. A. Pisner, and D. M. Schnyer (2020). Support vector machine, Editor(s): Mechelli, A. and Vieira, S. *Machine Learning*, pp. 101-121, Academic Press.
- [22] R. Gandhi (2018). Support Vector Machine — Introduction to machine learning algorithm. Available at <https://towardsdatascience.com/support-vector-machine-introduction-to-machine-learning-algorithm-934a444fca47>. Accessed on 15<sup>th</sup> May 2021.